

Data Comparison between Oracle and Postgres Using DB Solo

Created by Singh, Priti, last modified on Jan 25, 2022

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Overview

During Oracle to Aurora Migration, both the databases need to be kept in sync until the Application cut-over happens. (*[QBO is evaluating DB Solo for this purpose](#)*)

DB Solo supports data comparison among all major relational database products available today. It runs on multiple operating system platforms - *Linux, MAC, Windows, Solaris*

DB Solo Data Comparison Reports created on one machine with a specific operating system can be copied and run on other machines with any operating systems listed above.

DB Solo Installation

For **Windows** and **Mac** it is straight forward as shown in the links below -

Download Software - <https://www.dbsolo.com/download.html>

Installation- <https://www.dbsolo.com/help/general.html>

For **Linux** -

1. Save this file https://www.dbsolo.com/download/dbsolo_nojre.sh on linux under /tmp

2. Install Java 8

```
sudo yum install java-1.8.0

java -version
```

3. Install DB Solo

[Last version tested and used for QBO is **Version : 5.3.1 (Built : May 30, 2021)**]

```
cd /tmp

./dbsolo_nojre.sh
```

4. Provide inputs else installer will take default values

```
Starting Installer ...
This will install DB Solo 5 on your computer.
OK [o, Enter], Cancel [c]

Click Next to continue, or Cancel to exit Setup.
Please read the following License Agreement. You must accept the terms of
this agreement before continuing with the installation.

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```

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Last update: 03/22/2007

I accept the agreement

Yes [1], No [2]

1

Select the folder where you would like DB Solo 5 to be installed, then click Next.

Where should DB Solo 5 be installed?

[/home/oracle/dbsolo5]

/u01/ogg/scripts/dbsolo5

Create symlinks?

Yes [y, Enter], No [n]

y

Select the folder where you would like DB Solo 5 to create symlinks, then click Next.

[/usr/local/bin]

Extracting files ...

```
Create a desktop icon?  
Yes [y, Enter], No [n]  
Y  
Setup has finished installing DB Solo 5 on your computer.  
Finishing installation ...
```

DB Connections Setup

As X11 forwarding will not be possible or easy for ec2 Linux instances in AWS, we will create Connections and Reports on local Windows/Mac machines and then copy them to ec2 instance where DB Solo is installed. We can modify both on Linux manually if required.

Note:

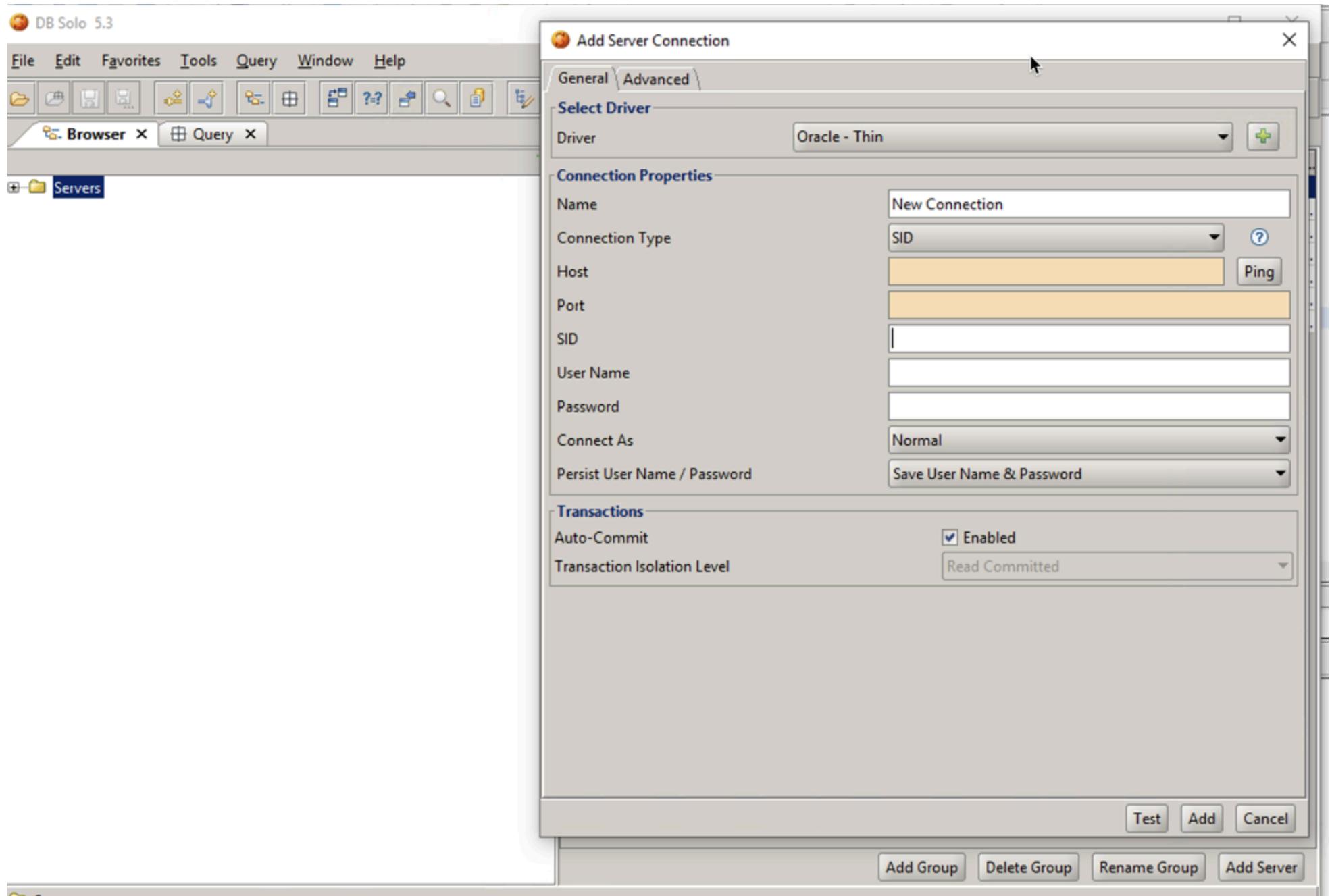
1. For using your local, you need to have a tunnel open to Oracle and Postgres/Aurora RDS instances.
2. Later when you copy the connections file to DB Solo ec2 instance, you may have to change the host and port number if not same.

You can refer to below link for general connection adding step.

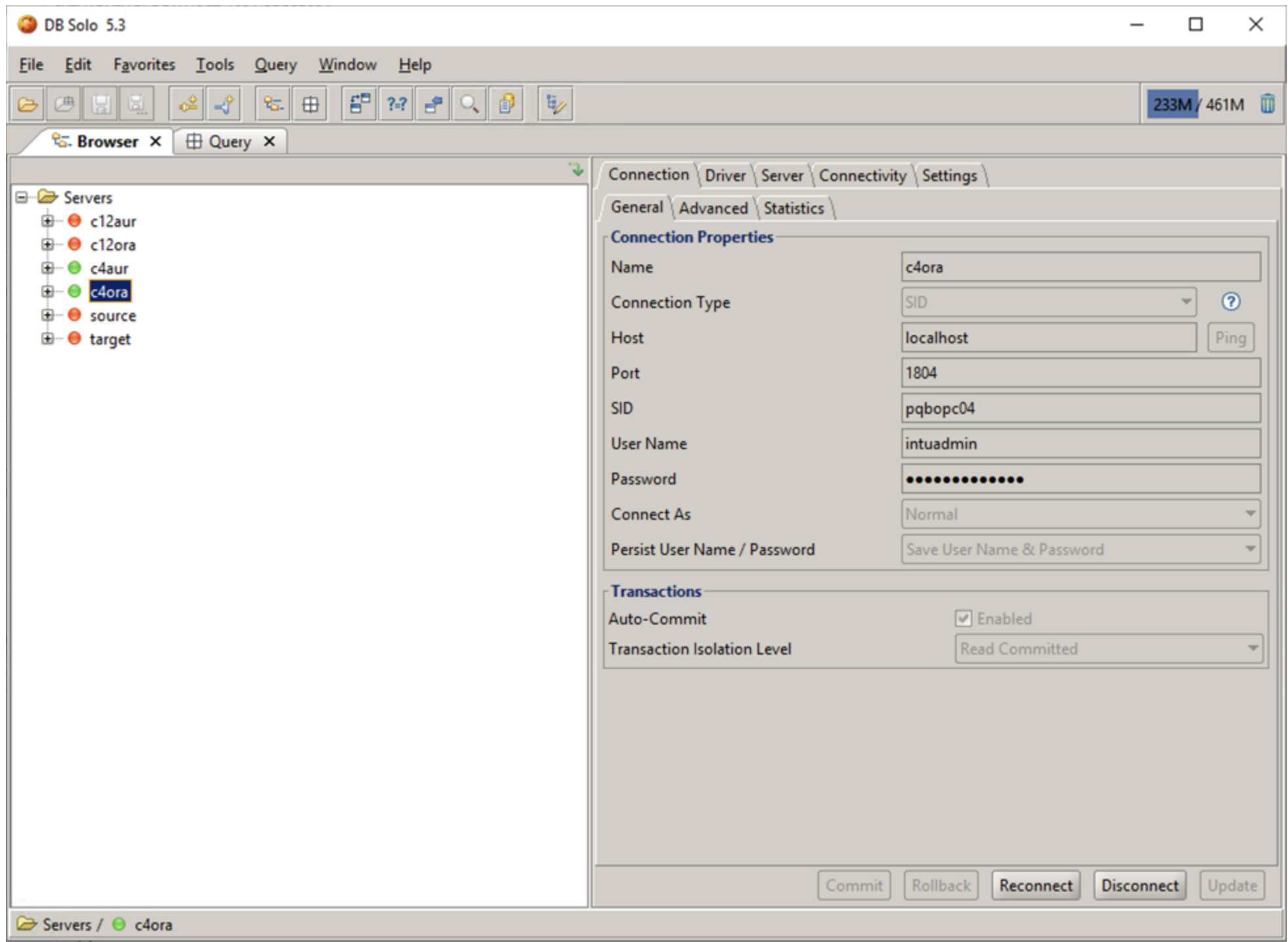
<https://www.dbsolo.com/help/connecting.html>

For QBO, from my local with tunneling-

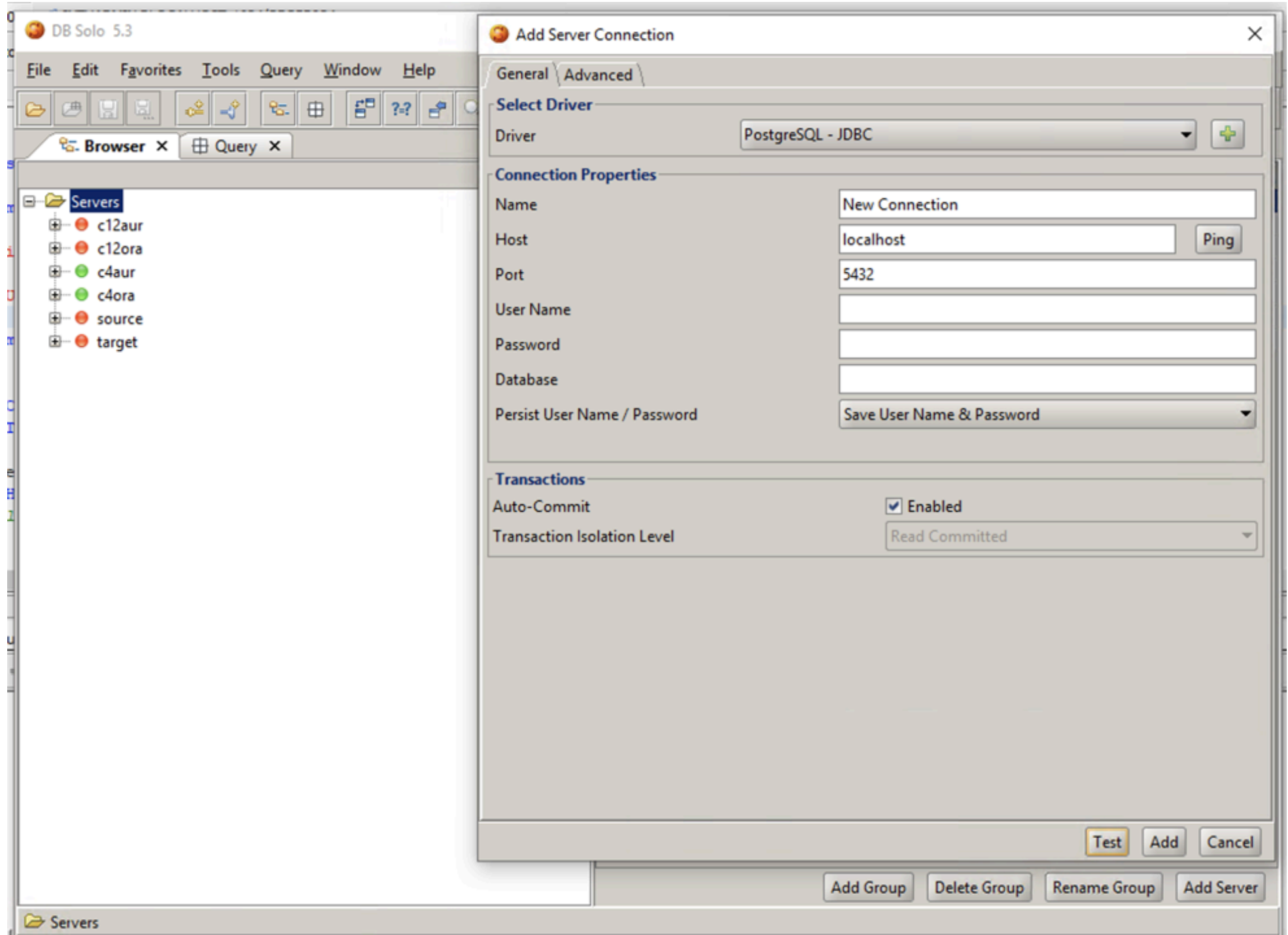
Source - Oracle



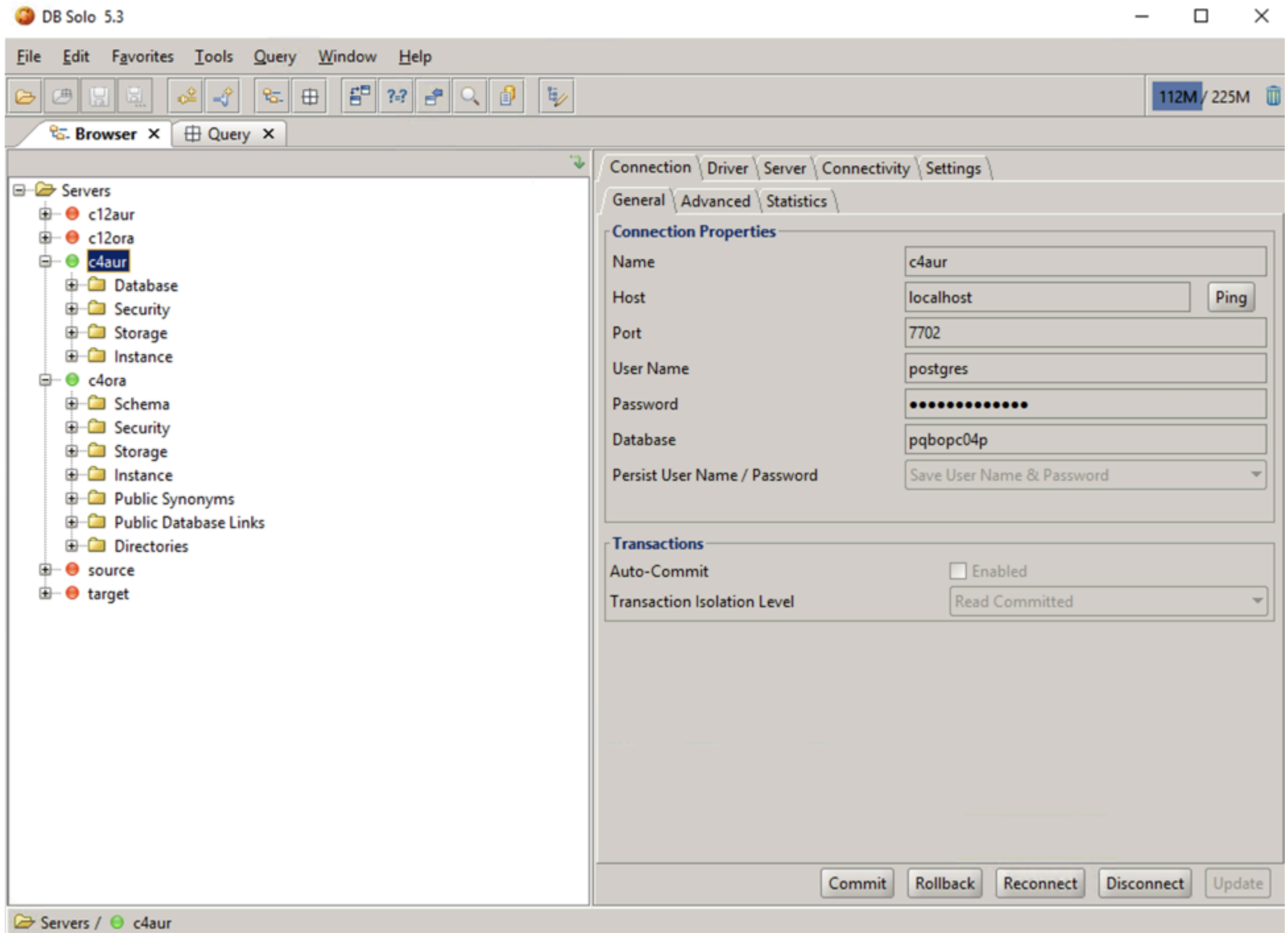
example



Target - Postgres



example

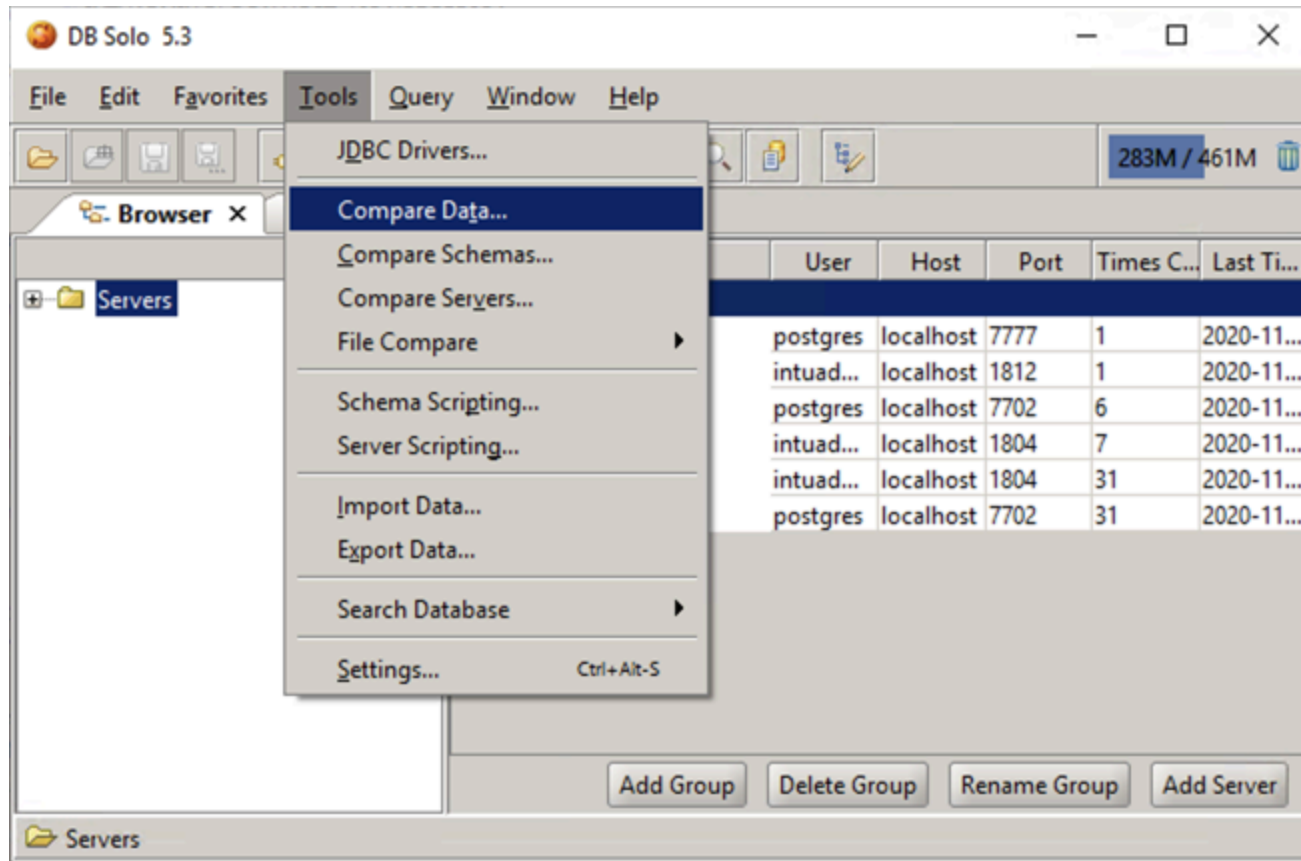


Data Comparison Report Setup

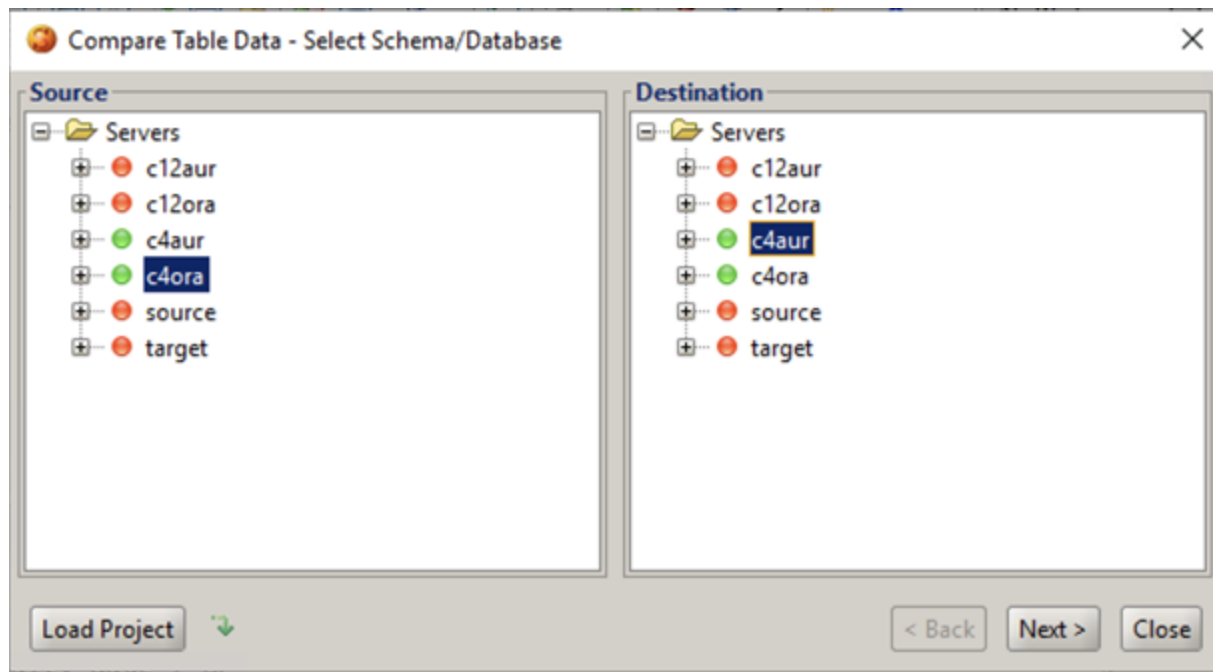
<https://www.dbsolo.com/help/datacomp.html>

For QBO -

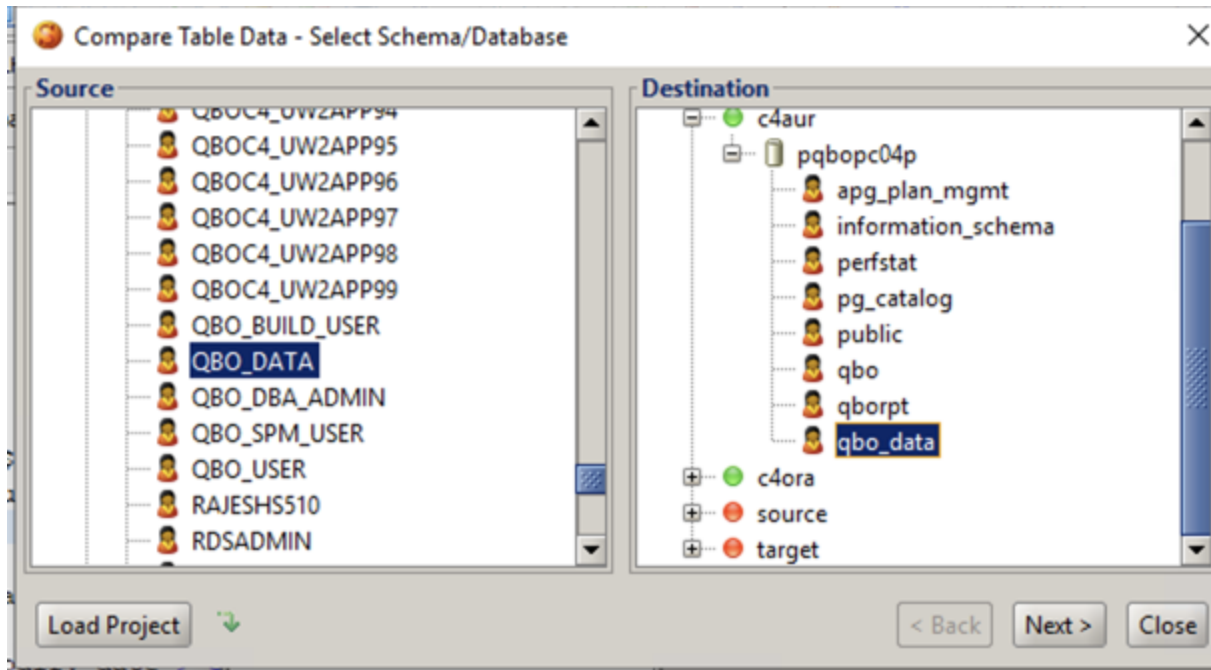
1. Select Compare Data option



2. Select Source and Target



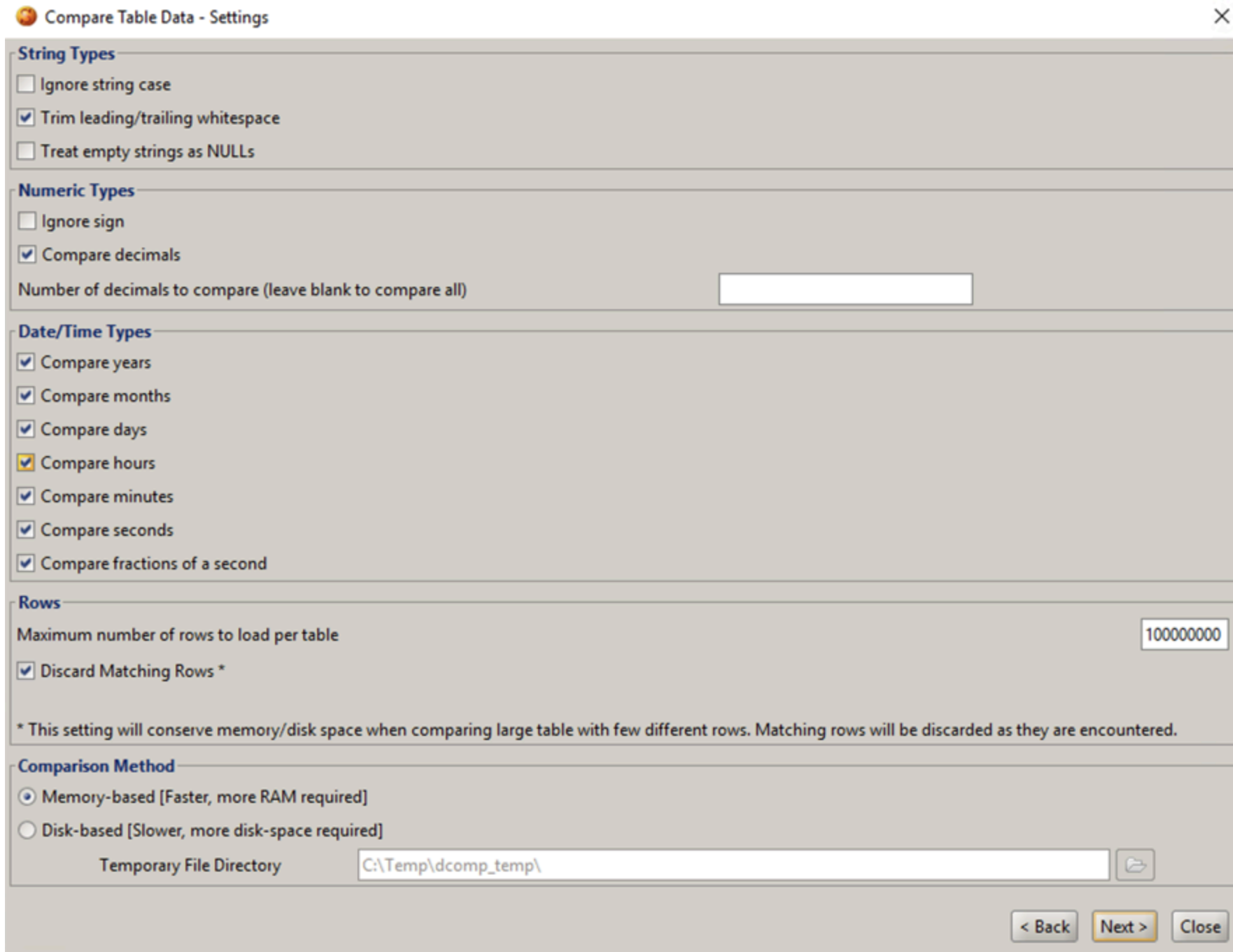
3. select Schema for Oracle and database/Schema for Postgres



4. (**Very Important**) make sure you have the right settings.

Note:

- Specify the MAX number of rows to be compared and, select the option to discard matching rows to show up in Report. It will take too much memory and file will be too big.
- Select *Memory-based* or *Disk-based* comparison method. Disk-based is very slow but works if you don't have much RAM



The dialog box is titled "Compare Table Data - Settings" and contains several sections for configuring the data comparison process. The "String Types" section has three checkboxes: "Ignore string case" (unchecked), "Trim leading/trailing whitespace" (checked), and "Treat empty strings as NULLs" (unchecked). The "Numeric Types" section has two checkboxes: "Ignore sign" (unchecked) and "Compare decimals" (checked), followed by a text input field for "Number of decimals to compare (leave blank to compare all)". The "Date/Time Types" section has six checkboxes, all of which are checked: "Compare years", "Compare months", "Compare days", "Compare hours", "Compare minutes", and "Compare seconds". The "Rows" section has a text input field for "Maximum number of rows to load per table" set to "100000000" and a checked checkbox for "Discard Matching Rows *". A note below states: "* This setting will conserve memory/disk space when comparing large table with few different rows. Matching rows will be discarded as they are encountered." The "Comparison Method" section has two radio buttons: "Memory-based [Faster, more RAM required]" (selected) and "Disk-based [Slower, more disk-space required]". Below this is a text input field for "Temporary File Directory" set to "C:\Temp\dcomp_temp\" with a browse button. At the bottom right are three buttons: "< Back", "Next >" (highlighted), and "Close".

String Types

- ☐ Ignore string case
- ☒ Trim leading/trailing whitespace
- ☐ Treat empty strings as NULLs

Numeric Types

- ☐ Ignore sign
- ☒ Compare decimals

Number of decimals to compare (leave blank to compare all)

Date/Time Types

- ☒ Compare years
- ☒ Compare months
- ☒ Compare days
- ☒ Compare hours
- ☒ Compare minutes
- ☒ Compare seconds
- ☒ Compare fractions of a second

Rows

Maximum number of rows to load per table

☒ Discard Matching Rows *

* This setting will conserve memory/disk space when comparing large table with few different rows. Matching rows will be discarded as they are encountered.

Comparison Method

- ☒ Memory-based [Faster, more RAM required]
- ☐ Disk-based [Slower, more disk-space required]

Temporary File Directory

< Back Next > Close

5. select the required tables

Compare Table Data - Select Tables

Select Tables / Columns

- ☐ CUSTOMERINTERVALS_1
- ☐ CUSTOMERS_1
- ☐ CUSTOMERTYPES_1
- ☐ CUSTOMFIELDDEFINITIONS_1
- ☐ CUSTOMFIELDDEFS_1
- ☐ CUSTOMFIELDIDENTITYMAPPINGS_1
- ☐ CUSTOMFIELDVALS_1
- ☐ DATAEXTDEFS_1
- ☐ DATAEXTS_1
- ☒ **DEPTS_1**
- ☐ DISCOUNTRATES_1
- ☐ DISCOUNTS_1
- ☐ EMAILDELIVERYINFO_1
- ☐ EMPLOYEECOMPENSATIONMAP_1
- ☐ EMPLOYEES_1
- ☐ ENTITYPREFS_1
- ☐ EVENTSTATUS_1
- ☐ EXPERIMENTS_1
- ☐ EXTERNALAUDITINFO_1
- ☐ EXTERNALTXREFERENCES_1
- ☐ EXTERNALTXREFERENCES_2
- ☐ FEATURELOCKS_1
- ☐ FEATURELOCKS_2
- ☐ FILEDTAXRETURNS_1

General **WHERE**

Name:

Type:

Schema:

Server:

Mapped Table/View:

Primary Key Columns

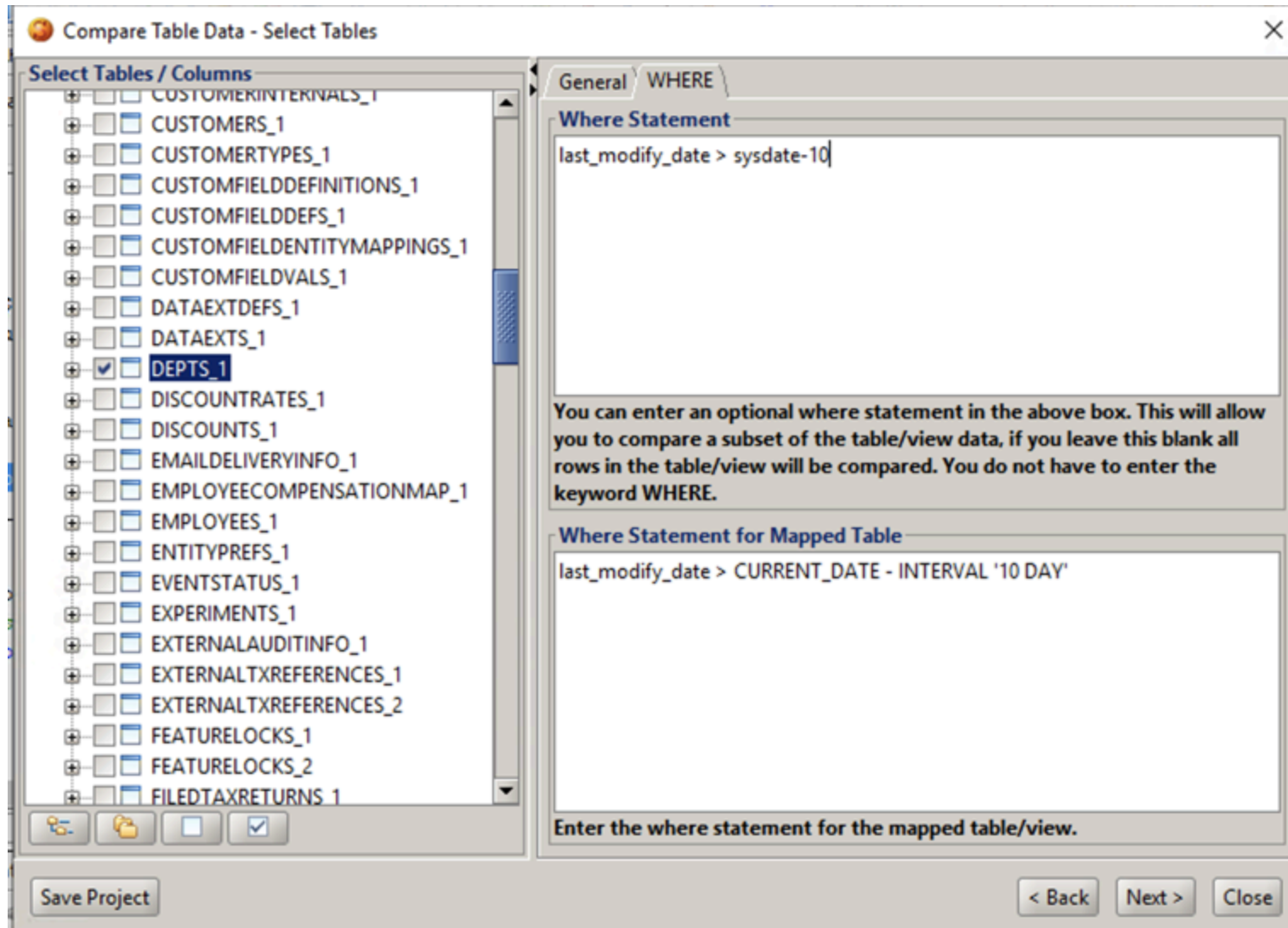
Column	Primary Key Column	Mapped Column
COMPANY_ID	☒	company_id
DEPT_ID	☒	dept_id
DEPT_NAME	☐	
PARENT_ID	☐	
NUM_CHILDREN	☐	
FULL_NAME	☐	
ORDER_BY_NAME	☐	
SUBLEVEL	☐	
HIDDEN	☐	
MERGED	☐	
OR_CUSTOM_STMT_TL	☐	

Primary key column order [optional] ?

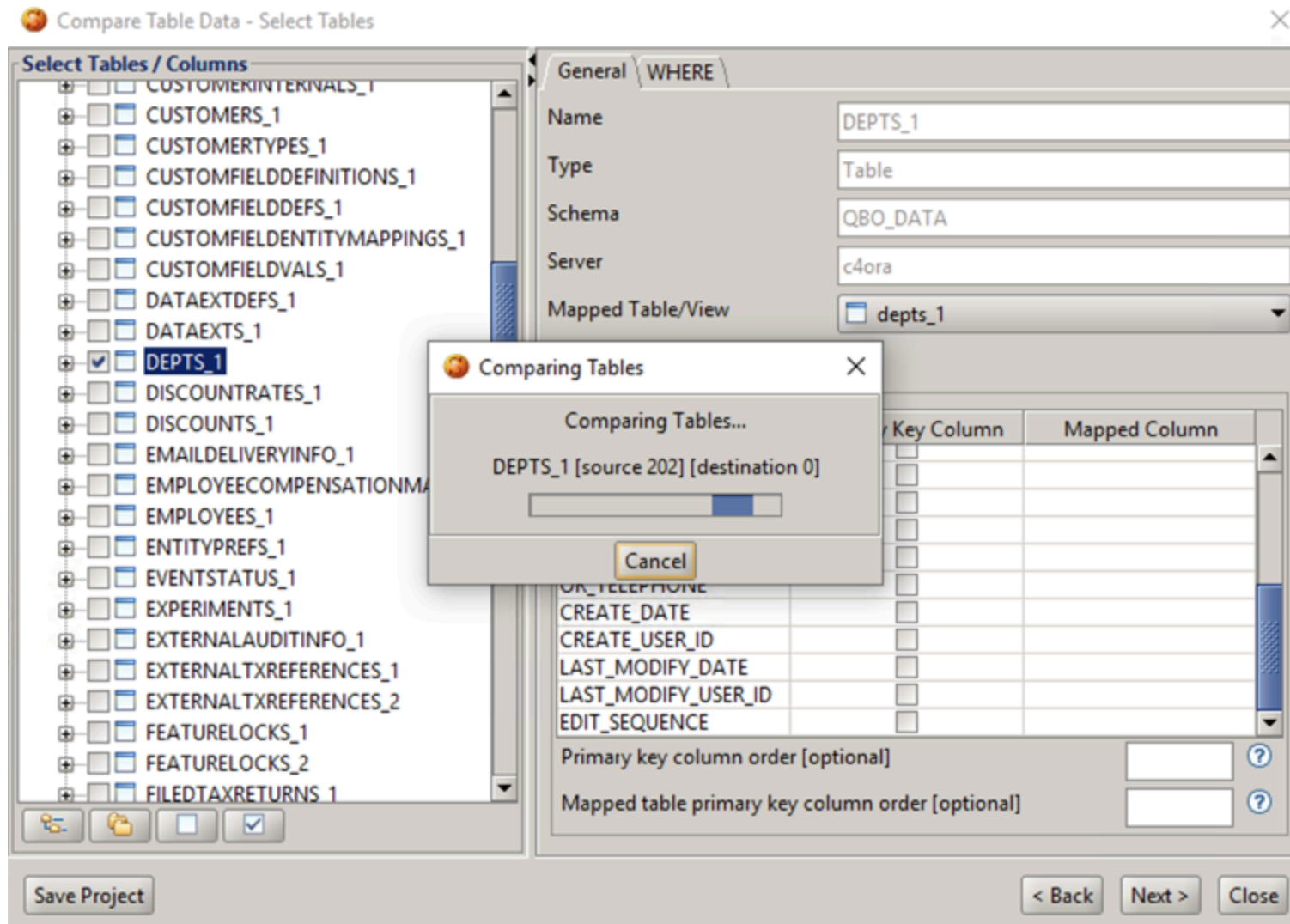
Mapped table primary key column order [optional] ?

Save Project **< Back** **Next >** **Close**

6. Specify the WHERE condition if needed (Syntax should be as per the database type)



7. Run the comparison and review the result



Compare Table Data - Results [c4ora/QBO_DATA -> c4aur/pqbopc04p/qbo_data] X

Table List

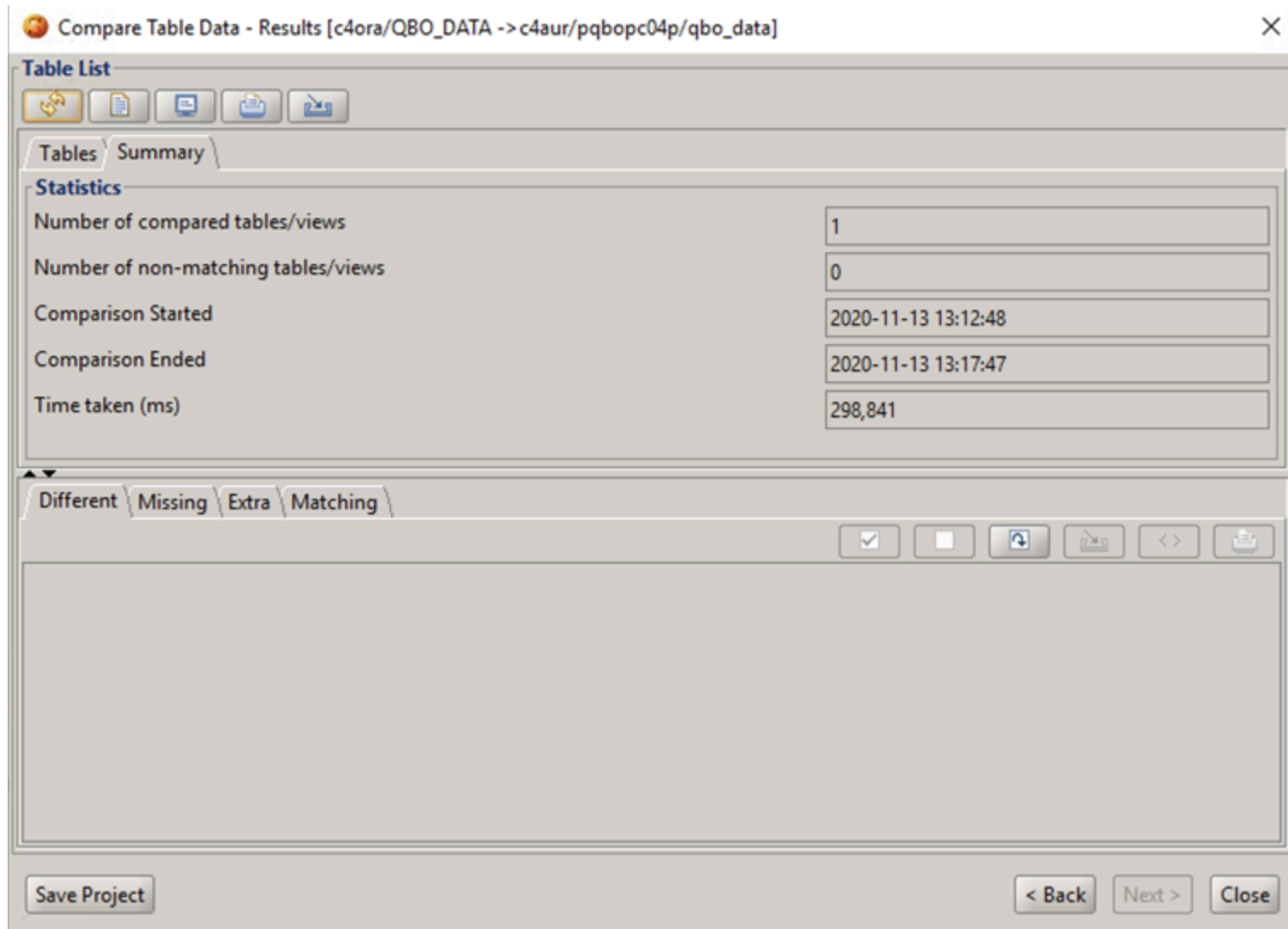
Tables Summary

Sync	Table	Rows	Matching	Different	Missing	Extra
☐	DEPTS_1	68923	68923	0	0	0

Select All Deselect All

Different Missing Extra Matching

Save Project < Back Next > Close



Generate xml for Comparison Reports

1. In the Compare Table dialogue box, there is a "Save Project" button in the bottom, Click it.

Compare Table Data - Select Tables

Select Tables / Columns

- ☐ CUSTOMERINTERVALS_1
- ☐ CUSTOMERS_1
- ☐ CUSTOMERTYPES_1
- ☐ CUSTOMFIELDDEFINITIONS_1
- ☐ CUSTOMFIELDDEFS_1
- ☐ CUSTOMFIELDIDENTITYMAPPINGS_1
- ☐ CUSTOMFIELDVALS_1
- ☐ DATAEXTDEFS_1
- ☐ DATAEXTS_1
- ☒ **DEPTS_1**
- ☐ DISCOUNTRATES_1
- ☐ DISCOUNTS_1
- ☐ EMAILDELIVERYINFO_1
- ☐ EMPLOYEECOMPENSATIONMAP_1
- ☐ EMPLOYEES_1
- ☐ ENTITYPREFS_1
- ☐ EVENTSTATUS_1
- ☐ EXPERIMENTS_1
- ☐ EXTERNALAUDITINFO_1
- ☐ EXTERNALTXREFERENCES_1
- ☐ EXTERNALTXREFERENCES_2
- ☐ FEATURELOCKS_1
- ☐ FEATURELOCKS_2
- ☐ FILEDTAXRETURNS_1

General **WHERE**

Name:

Type:

Schema:

Server:

Mapped Table/View:

Primary Key Columns

Column	Primary Key Column	Mapped Column
COMPANY_ID	☒	company_id
DEPT_ID	☒	dept_id
DEPT_NAME	☐	
PARENT_ID	☐	
NUM_CHILDREN	☐	
FULL_NAME	☐	
ORDER_BY_NAME	☐	
SUBLEVEL	☐	
HIDDEN	☐	
MERGED	☐	
OR_CUSTOM_STMT_TL	☐	

Primary key column order [optional]

Mapped table primary key column order [optional]

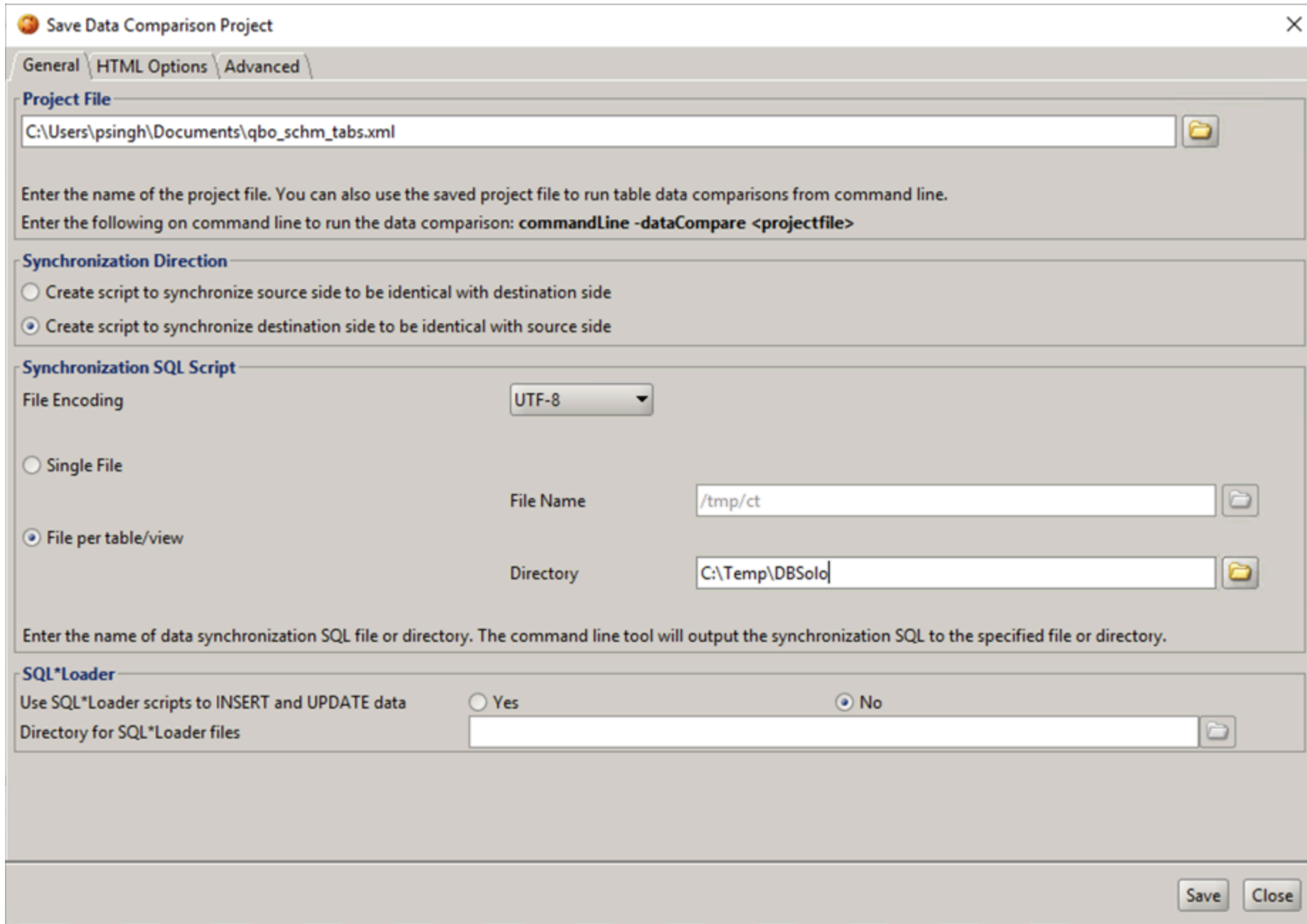
Save Project **< Back** **Next >** **Close**

2. Specify the path where you want to generate the xml file.

Note:

There is an option to generate synchronization script for mismatches. You can generate one file for all or one file per table.

If you are not going to use this synchronization script then its useless to generate it. It takes space. On windows, you get error if you leave it blank. Luckily on Linux, you can leave it blank.



The dialog box is titled "Save Data Comparison Project" and has a close button (X) in the top right corner. It contains three tabs: "General", "HTML Options", and "Advanced". The "General" tab is selected and contains the following sections:

- Project File**: A text field containing "C:\Users\psingh\Documents\qbo_schm_tabs.xml" and a folder icon button. Below it, a text area contains the instruction: "Enter the name of the project file. You can also use the saved project file to run table data comparisons from command line. Enter the following on command line to run the data comparison: **commandLine -dataCompare <projectfile>**".
- Synchronization Direction**: Two radio buttons. The first is "Create script to synchronize source side to be identical with destination side". The second is selected: "Create script to synchronize destination side to be identical with source side".
- Synchronization SQL Script**: A section with a "File Encoding" dropdown set to "UTF-8". Below it are two radio buttons: "Single File" and "File per table/view" (which is selected). To the right of these are two text fields: "File Name" with the value "/tmp/ct" and a folder icon button, and "Directory" with the value "C:\Temp\DBSolo" and a folder icon button. Below these fields is a text area with the instruction: "Enter the name of data synchronization SQL file or directory. The command line tool will output the synchronization SQL to the specified file or directory."
- SQL*Loader**: A section with the text "Use SQL*Loader scripts to INSERT and UPDATE data" followed by two radio buttons: "Yes" and "No" (which is selected). Below this is a text field for "Directory for SQL*Loader files" and a folder icon button.

At the bottom right of the dialog box are "Save" and "Close" buttons.

3. Specify where you want to see the final Data comparison HTML report file.

 Save Data Comparison Project ✕

General

HTML Options

Advanced

Output File
c:\temp\qbo_schm_tab.html 

Output Options

General

Page Title

DB Solo - Data Compare Results

Heading

DB Solo - Data Compare Results

☐ Include matching tables in report

☒ Generate detail page for each table

☐ Include matching rows on detail page

☒ Include timestamp in detail page file name

Header Row

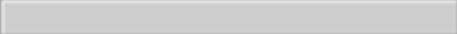
Font Size

11

Column Alignment

Left

Background Color



Change

Even Rows

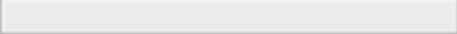
Font Size

11

Column Alignment

Left

Background Color



Change

Odd Rows

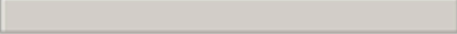
Font Size

11

Column Alignment

Left

Background Color



Change

Non-Matching Columns [Table Detail Page]

Font Size

11

Column Alignment

Left

Background Color



Change

Save

Close

Copy Connections and Comparison Reports for other environments

1. Comparison Report XML will be generated in the path you specified in the previous step.
i.e., `C:\Users\psingh\Documents\qbo_schm_tabs.xml`
2. Saved Connections will be found under your USER profile home directory in **.dbsolo5** folder. File name will be **connections.xml**
i.e., `C:\Users\psingh\.dbsolo5\connections.xml`

Copy both the files to the required DB Solo machine (in QBO case, it's ec2 linux) .

Note: You can copy Report xml anywhere you want. **connections.xml** must go to USER's profile home directory. For example for user oracle, it should be under `\home\oracle\.dbsolo5\`

Run Data Comparison Report on Linux

Go to the installation directory (As per the installation step above, our DB Solo installation directory is `/u01/ogg/scripts/dbsolo5` and copied Report xml is `qbo_schm_tabs.xml`)

```
cd /u01/ogg/scripts/dbsolo5

./commandLine -dataCompare qbo_schm_tabs.xml

or

nohup ./commandLine -dataCompare qbo_schm_tabs.xml &
```

Note:

1. If you get any memory/heap errors, you can edit `commandLine` file to use more memory

```
cd /u01/ogg/scripts/dbsolo5

vi commandLine

-- Go to the below lines and update :
```

```
$INSTALL4J_JAVA_PREFIX exec "$app_java_home/bin/java" "-Xms2128m" "-Xmx134024m"
```

2. if you get memory/heap errors on Windows, you can launch dbsolo from Command prompt using the below -

```
dbsolo -J-Xmx14096m
```

Common Config Changes when running on Linux

a) In the Report xml and the connections.xml, you can modify the below **entries** to run the report for a different set of databases, say for c12 oracle and aurora databases-

You can of-course change user/password/port etc if needed.

--In Report xml file -

```
<source name="c4ora" id="1603919423072" schema="QBO" database=""/>
<destination name="c4aur" id="1604429110459" schema="qbo" database="pqbopc04p"/>
```

-- In connections.xml

```
<connection driver="Oracle - Thin" name="c4ora" persist_username="2" user="intuadmin" password="72F75B1" domain="" host="pqbopc04.sbg-qbo-ppd.a.intuit.com" url="" database="" port="1521" file="" sid="pqbopc04" servicename="" autocommit="false" isolation="2" identifiers="NONE" connectivity="false" reconnect="false" connectivity_interval="3" server_db_type="1" connect_as="0" connection_type="1">
</properties>
</connection>
<connection driver="PostgreSQL - JDBC" name="c4aur" persist_username="2" user="postgres" password="72F75B1" domain="" host="pqbopc04p.cluster-cozwmkqglitfx.us-west-2.rds.amazonaws.com" url="" database="pqbopc04p" port="5432" file="" sid="" servicename="" autocommit="false" isolation="2" identifiers="NONE" connectivity="false" reconnect="false" connectivity_interval="3" server_db_type="11" connect_as="0" connection_type="0">
</properties>
</connection>
```

Note: In connections.xml, you can add multiple connections, but report can use one connection at a time.

b) In setting.xml (**IMPORTANT**)

Change the logging level

Only critical errors - 1000

All info - 800 (**RECOMMENDED**)

c) In Report.xml -

Enable/Disable a table or column from comparison

DB Solo Limitations

- Compared to Veridata (which is only available for Oracle databases), DB Solo (that compares Oracle and Postgres) is little slow. Adding WHERE clause on date fields helps. ([DBSolo QBO/QBO_DATA Tables Comparison Report & Details](#))
- Veridata has an option to do server-side sorting instead of sorting in the database. DB Solo has no such option to offload sorting from database.
- DB Solo needs a PK or a set of KEYCOLs to uniquely identify every record to be compared. It will error out if any duplicate record.
- Keycol cannot have a column with JSON constraint. It errors out.
- Data Comparison happens in the memory of the DB Solo server. It logs the comparison result for all the tables in the end in an html file. If any failure at any step , no data will be written out. If comparison is for 100 tables and if any failure on the last table, no report will be generated even for the 99 tables got compared successfully.
- ~~ACCOUNTANTFIRMS_1 table comparison has the below error for empty/null blob columns - (fixed)~~
~~WARNING: UB: Error when loading column FIRM_LOGO, data type=BLOB, val=null, row=55796: null~~
~~java.lang.NegativeArraySizeException~~
- DBSolo pulls all tables data in first phase and then in second phase does the comparison. It needs lot of memory& storage

References

<https://www.dbsolo.com/index.html>

No labels

